Environmental Acoustic Mapping of High-Density Urban Micro-Clusters in Delhi Using GIS-Integrated Calibrated Smartphone Monitoring: Spatial Hotspot Identification and Exceedance Analysis

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Abstract

Environmental noise is a widely under-monitored urban health hazard in Indian cities. Delhi, with over 32 million people and more than 12 million registered vehicles, sits at the extreme end of the problem, and dense, mixed-use localities such as Jamia Nagar remain largely unmapped at the neighbourhood scale. This study measured and spatially mapped environmental noise across three high-density micro-clusters in Jamia Nagar — Batla House (commercial), Gaffar Manzil (residential), and Okhla Head (commercial, arterial corridor) — during four monitoring sessions, and compared recorded levels against Central Pollution Control Board (CPCB) and World Health Organization (WHO) noise standards. A smartphone (Samsung Galaxy S7 Edge) running the Decibel X application was used for field measurements at 36 geo-tagged points, and was validated against a reference Sound Level Meter at CSIR–Central Road Research Institute (CRRI) prior to fieldwork using a 0–20,000 Hz frequency sweep (Pearson $r = 0.933$, $R^2 = 0.9388$, RMSE = 1.39 dB(A), MAE = 1.94 dB(A)). Spatial noise maps were generated using Inverse Distance Weighting (IDW) interpolation in QGIS. CPCB limits were exceeded at every location during every monitoring period. Batla House (commercial) recorded mean working day evening equivalent continuous sound level (Leq) of 75.5 dB(A) against a 65 dB(A) commercial daytime limit (+10.5 dB(A)), with maximum instantaneous level (Lmax) peaking at 102.6 dB(A). Gaffar Manzil (residential) exceeded its 55 dB(A) daytime limit by up to 11.3 dB(A). At midnight, all three clusters exceeded both the applicable CPCB nighttime limit and the WHO Lnight guideline of 45 dB(A). GIS maps identified commercial frontages and arterial road segments as the dominant acoustic hotspots. The calibrated smartphone–GIS workflow proved practical and reliable for neighbourhood-scale acoustic mapping, providing a direct evidence base for traffic management, noise governance, and urban planning decisions in high-density Indian urban localities.

Keywords: *urban noise mapping; geographic information system; IDW interpolation; smartphone monitoring; Decibel X; equivalent continuous sound level; CPCB; Delhi; acoustic hotspot; mixed land-use*

1. Introduction

Noise is constant in cities like Delhi traffic, horns, vendors, construction, and community activity layered on top of each other without pause. What makes it dangerous is precisely that constancy. Environmental noise is classified as the second most harmful environmental stressor after air pollution, with chronic exposure linked to cardiovascular disease, sleep disruption, cognitive impairment, and reduced quality of life [1].

India's regulatory framework exists on paper. The Noise Pollution (Regulation and Control) Rules [3] set daytime limits of 55 dB(A) for residential areas and 65 dB(A) for commercial areas, with nighttime limits of 45 and 55 dB(A) respectively [2]. WHO recommends Lnight below 45 dB(A) for sleep protection [1]. But enforcement is weak, monitoring is sparse, and detailed spatial data for specific high-density neighbourhoods is almost entirely absent from the published literature [4].

Jamia Nagar in South-East Delhi is a case that deserves attention. Population density in its core clusters exceeds 40,000 persons per square kilometre. Land use is a dense mix of residential, commercial, religious, and educational functions, often within the same building, and neighbourhood-level noise monitoring has not been done systematically.

This study addresses that gap directly. Field-based noise monitoring was conducted at 36 geo-tagged points across three distinct clusters within Jamia Nagar Batla House, Gaffar Manzil, and Okhla Head during four temporally distinct sessions covering weekday morning, weekday evening, Sunday evening, and midnight conditions. Measurements were made using a smartphone running the Decibel X application, validated against a reference Sound Level Meter at CSIR–CRRI. Spatial noise maps were produced using IDW interpolation in QGIS.

1.1 Literature background

Research across Indian cities consistently shows recorded noise levels exceeding CPCB limits, with traffic identified as the dominant cause. Pathak et al. [5] monitored ten residential sites in Lucknow and found Leq values between 70 and 84 dB(A). Banerjee [6] surveyed 30 sites in Asansol and confirmed that land-use type shapes acoustic exposure. Singh and Davar [7] identified traffic and industrial activity as the two dominant noise sources across Indian cities.

Studies specific to Delhi reinforce this pattern. Chakraborty et al. [8] monitored 135 locations city-wide and found CPCB exceedances in every zone. Chauhan et al. [9] extended this to 100 sites across the National Capital Territory and reported Leq values predominantly in the 70–80 dB(A) range. Kumar et al. [10] found commercial zones in Haridwar recording the highest noise due to traffic-tourism interactions, mirroring conditions observed in Batla House in the present study. Mahesh et al. [11] documented weekday-to-weekend reductions of 3–5 dB(A) in Mysuru commercial zones, a contrast that proved much smaller at Batla House in this study. Das et al. [12] combined noise mapping with a population vulnerability assessment in a meso-level urban environment of Eastern India, an approach pairing acoustic exposure with population sensitivity. Devasia et al. [13] applied GIS-aided noise pollution mapping in urban Puducherry, demonstrating the applicability of the GIS-IDW approach beyond major metro contexts.

GIS-based spatial interpolation is the standard approach for visualising urban noise distribution. IDW interpolation in particular is widely applied for its computational efficiency and suitability for moderately dense point datasets. Garg and Maji [14] applied GIS-IDW in Jalandhar and found exceedances in all zones. Mehta and Bhuva [15] mapped noise in Surat using IDW at 20 locations and identified high-noise road corridors. Kalita et al. [16] used GIS-IDW at 23 sites in Guwahati and delineated hotspots at traffic intersections. Garg et al. [17] showed GIS-predicted noise levels in Patiala matched observed values closely. Internationally, Cai et al. [18] used GIS with the RLS-90 model in Macao and estimated over half the population exposed above 65 dB(A); Haq and El-Gendy [19] mapped 80 sites in Dammam and found widespread WHO guideline exceedances; and Aumond et al. [20] documented significant spatiotemporal noise variability in dense Paris neighbourhoods. Murphy and King [21] have linked GIS noise exposure data directly to cardiovascular and psychological health outcomes.

A separate methodological strand is relevant here. Zuo et al. [22] demonstrated a region-based noise-mapping approach in which sampling points are tagged by surrounding land-use type and interpolated separately within each type before being combined, arguing that standard interpolation methods such as IDW and ordinary Kriging do not account for the spatial heterogeneity that arises when noise behaves differently across adjacent land-use types at short distances. This is directly relevant to clusters such as Gaffar Manzil, where a single IDW surface must represent both a busy access road and a quiet residential interior; this heterogeneity issue is revisited in Section 6.

The scientific case for calibrated smartphone-based noise monitoring is well established. Kardous and Shaw [23] evaluated multiple smartphone applications and found several performing within ±2 dB(A) of reference SLMs when properly calibrated. Murphy and King [24] tested 100 smartphones across 1472 measurements in a reverberation room and found that calibrated applications could approach reference SLM accuracy. Nast et al. [25] showed carefully calibrated apps produce A-weighted levels comparable to Type 2 SLMs. This study extends those validation protocols and applies the smartphone–GIS combination to a neighbourhood-scale monitoring exercise that has not previously been conducted in this part of Delhi.

1.2 Research gap

Most Delhi noise studies have focused on arterial roads or institutional zones. The internal acoustic dynamics of dense mixed-use residential-commercial clusters the kind of places where hundreds of thousands of people actually live have received almost no attention. There is also a methodological gap: calibrated smartphone-based GIS noise mapping has not been applied at the micro-cluster scale in Indian metropolitan settings. This study addresses both gaps.

1.3 Novelty and contribution of this study

The novelty of this study can be summarised as follows:

(1) a formally validated, calibrated smartphone-based monitoring protocol (Decibel X), benchmarked against a reference Sound Level Meter at CSIR–CRRI, is applied at neighbourhood micro-cluster scale rather than city-wide arterial-road scale;

(2) three urban micro-clusters with distinct zoning and morphology (commercial corridor, residential interior, arterial road corridor) are monitored under an identical four-session temporal design, enabling direct cross-cluster and cross-period comparison;

(3) exceedance analysis is applied using zone-appropriate CPCB limits (commercial vs. residential, day vs. night) rather than a single uniform threshold, providing a more regulation-accurate assessment of compliance; and

(4) the GIS-IDW workflow and its limitations under sparse point density are explicitly examined, with reference to region-stratified interpolation alternatives for future application.

1.4 Objectives

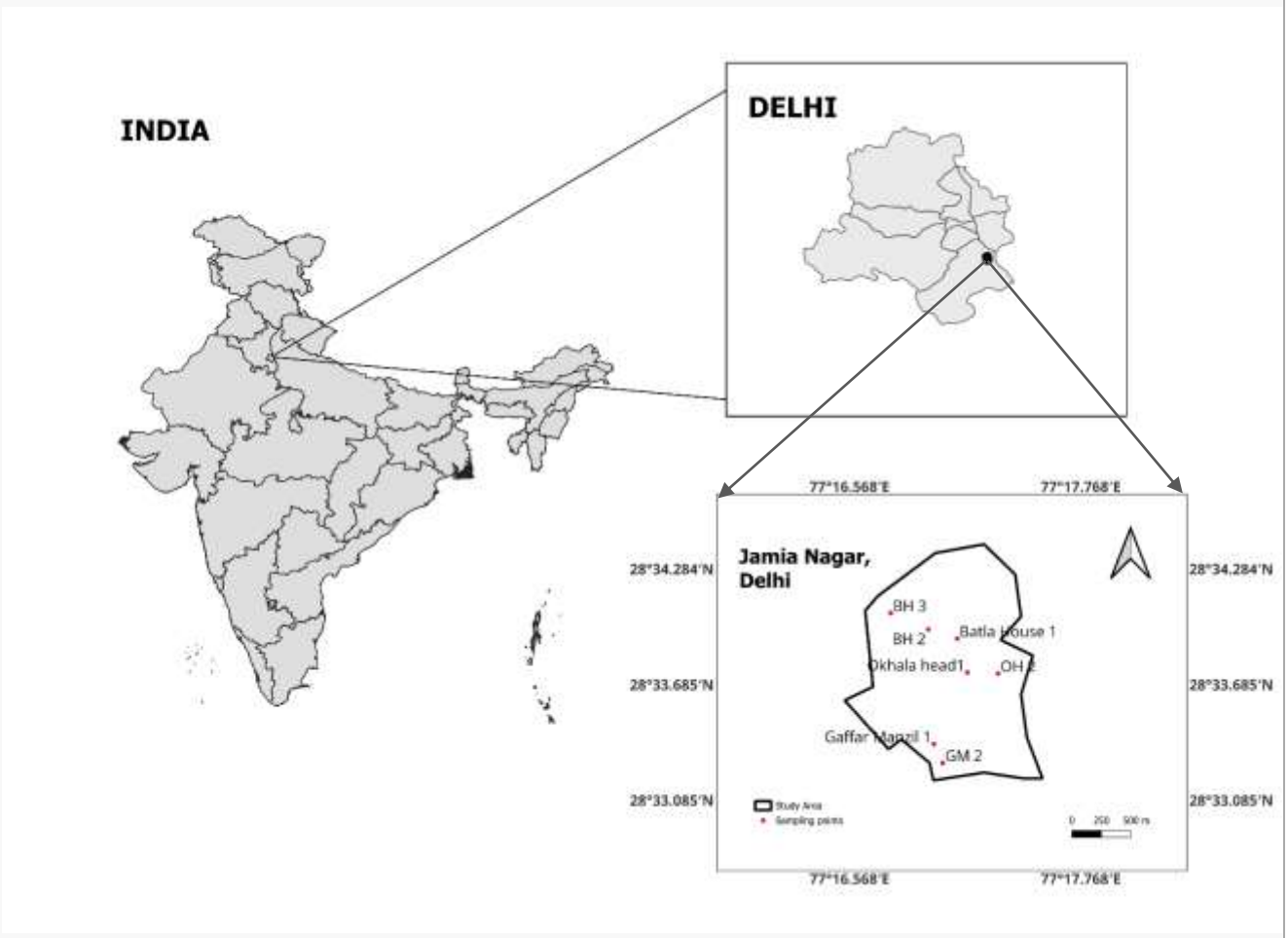
- To conduct calibrated, geo-tagged field noise monitoring at 36 points across Batla House, Gaffar Manzil, and Okhla Head during four monitoring sessions.
- To validate a smartphone-based monitoring device (Decibel X) against a reference Sound Level Meter at CSIR–CRRI before fieldwork.
- To generate IDW spatial interpolation noise maps (Leq and Lmax) for each cluster and session using QGIS.
- To compare all recorded levels against the applicable CPCB Noise Rules and WHO Environmental Noise Guidelines, and quantify exceedances.
- To identify spatial acoustic hotspots and relate them to land use, traffic, and urban layout.
- To assess public health implications and propose practical policy recommendations.

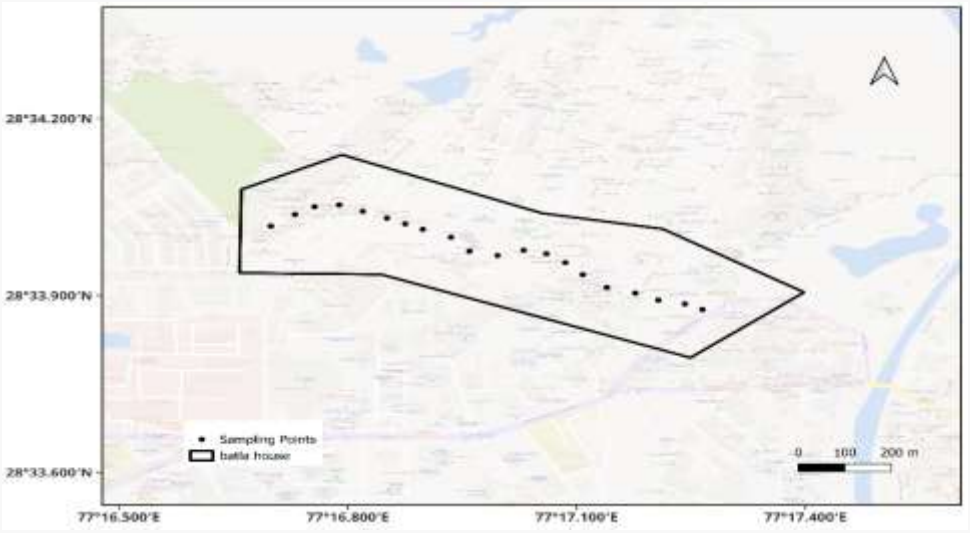
2. Study area

2.1 Jamia Nagar, South-East Delhi

Jamia Nagar sits in the South-East District of Delhi, roughly bounded by Mathura Road to the north, the Yamuna floodplain to the east, Shaheen Bagh to the south, and the Jamia Millia Islamia University campus to the north-west. The locality covers around 10 km² and is home to between 400,000 and 600,000 people. Core cluster densities exceed 40,000 persons per km². Major roads include Okhla Road, Abul Fazal Road, Zakir Husain Marg, and Batla House Road.

Land use is genuinely mixed in a way that planning categories rarely capture. Ground floors of residential buildings routinely operate as shops, food stalls, or service centres. Mosques, schools, hospitals, and markets share the same narrow streets, with no meaningful acoustic buffer between residential and commercial activities.

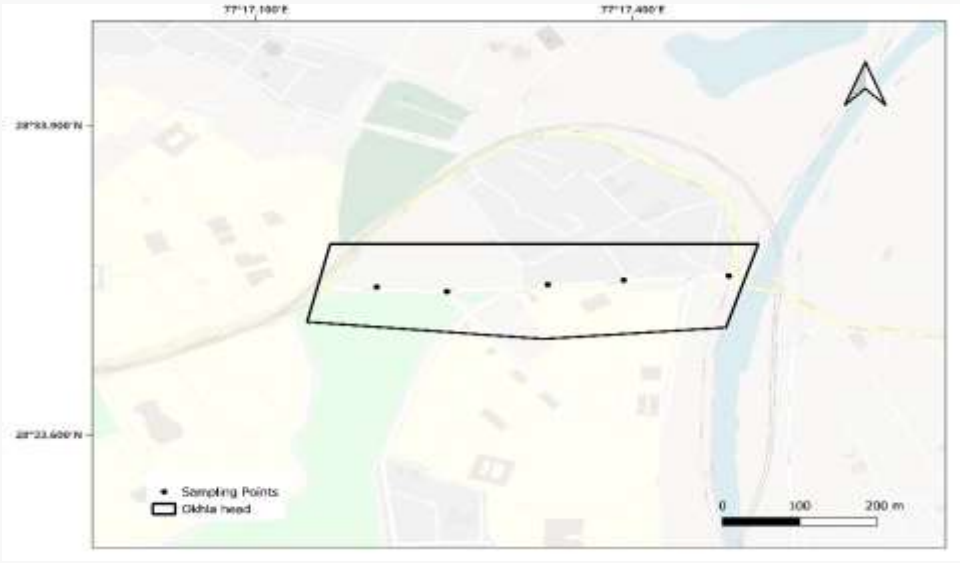




Batla House Sampling Points



Gaffar Manzil Sampling Points



Okhla Head Sampling Points

Figure 1. Location map showing Jamia Nagar within South-East Delhi, India, and the three study clusters (Batla House, Gaffar Manzil, and Okhla Head) with sampling point locations.

2.2 Batla House (commercial zone)

Batla House (approx. 28°33.9'–28°34.2'N; 77°16.8'–77°17.4'E) is classified under CPCB Noise Rules as a commercial zone and is the most commercially active of the three clusters. The main road carries a constant mix of private cars, two-wheelers, auto-rickshaws, cycle-rickshaws, goods vehicles, and buses. Road widths typically run 5–12 m but are frequently reduced by parked vehicles and street vendors, with three- to five-storey buildings lining both sides of the narrow carriageways. Twenty sampling points (BH001–BH020) covered the main commercial corridor and adjacent side streets over roughly 600 m.

2.3 Gaffar Manzil (residential zone)

Gaffar Manzil (approx. 28°33.24'–28°33.6'N; 77°16.9'–77°17.2'E) is classified as a residential zone. Its eastern edge lies close to Zakir Husain Marg, which brings traffic noise into the cluster. Interior lanes are narrow (3–6 m) and mostly inaccessible to large vehicles, though two-wheelers and auto-rickshaws move through freely. Eleven sampling points (GM001–GM011) covered the interior and the main access road.

2.4 Okhla Head (commercial zone, arterial corridor)

Okhla Head (approx. 28°33.6'–28°33.9'N; 77°17.1'–77°17.5'E) is classified as a commercial zone and is defined acoustically by Okhla Road, which carries heavy and continuous truck, bus, and goods-vehicle traffic even at night. Five sampling points (OH001–OH005) were placed along a transect from the road edge towards the residential hinterland.

3. Materials and methods

3.1 Instrumentation

Noise monitoring was carried out using a Samsung Galaxy S7 Edge smartphone running the Decibel X application, which records A-weighted equivalent continuous sound level (Leq) and maximum instantaneous sound level (Lmax) over a user-defined duration. A-weighting is the standard frequency weighting for environmental noise assessment, matching the sensitivity of the human ear across the audible frequency range, and the app records readings consistent with ISO 1996-1:2016 measurement requirements [26].

3.2 Validation of smartphone-based noise monitoring

Before field measurements were taken, a calibration exercise was conducted at CSIR–Central Road Research Institute (CRRI), New Delhi, under the supervision of Dr. Nasim Akhtar, Senior Principal Scientist, and Sakshi Gupta, Scientist/Technician. The smartphone (Decibel X) and Smartphone measurements obtained using the Decibel X application were calibrated against a Class 1 Integrating Sound Level Meter (SLM 109, Envirotech Instruments Pvt. Ltd., India) compliant with IEC 61672 standards placed on a single tripod, in the same camera frame, at 50 cm from a laptop speaker. A 0–20,000 Hz frequency sweep (a rising-tone audio recording) was played at maximum laptop volume while both instruments recorded simultaneously on video. Readings from both instruments were read off the recording at 2-second intervals, giving 23 paired data points spanning approximately 45–100 dB(A) the full range expected in the field.

The paired dataset was analysed statistically (Table 1). Pearson correlation was $r = 0.933$, coefficient of determination $R^2 = 0.9388$, root mean square error (RMSE) = 1.39 dB(A), and mean absolute error (MAE) = 1.94 dB(A). These values are within the ±2 dB(A) measurement uncertainty accepted for environmental noise monitoring under ISO 1996-1:2016 and CPCB guidelines, and are consistent with performance metrics reported by Kardous and Shaw [23], Murphy and King [24], and Nast et al. [25] for calibrated smartphone applications. This correlation also compares favourably with the device-suitability threshold implied by Zuo et al. [22], who found a smartphone-SLM correlation of 0.8951 to be lower than that of devices they considered suitable for environmental noise measurement (e.g. 0.9792 for a calibrated phone fitted with an external microphone); the present correlation of 0.933 falls well within the range associated with suitable devices in that study. The regression equation derived from the calibration session ($y = 0.4346x + 67.638$, where x is the raw Decibel X reading and y is the SLM-equivalent corrected value) was applied as a correction factor to all subsequent field measurements.

Table 1
Smartphone calibration validation results versus reference Sound Level Meter (CSIR–CRRI).

Metric	Value	Acceptable threshold
Pearson correlation (r)	0.933	≥0.90
Coefficient of determination (R^2)	0.9388	≥0.85
RMSE	1.39 dB(A)	≤2.0 dB(A) (ISO/CPCB)
MAE	1.94 dB(A)	≤2.0 dB(A)

Metric	Value	Acceptable threshold
Regression equation	y = 0.4346x + 67.638	Applied as correction factor

Figure 2. Calibration comparison between the Decibel X smartphone application (Samsung Galaxy S7 Edge) and the reference Sound Level Meter (CSIR–CRRI). The trendline equation is $y = 0.4346x + 67.638$, with $R^2 = 0.9388$.

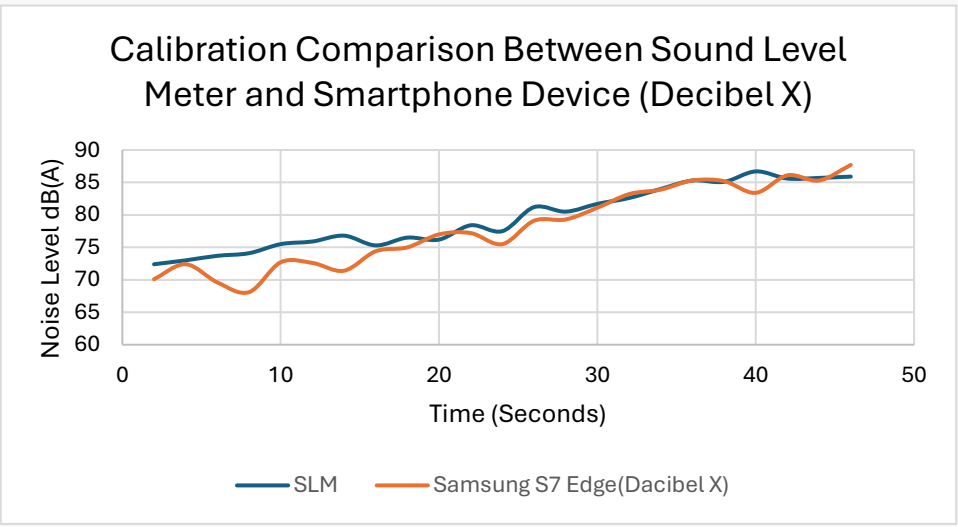


Fig. 2 — Calibration graph: 23 paired points, regression line and equation.

3.3 Sampling design

Sampling points were selected to give spatial coverage of each cluster across different land-use types and traffic conditions: 20 points (BH001–BH020) in Batla House along the main commercial corridor and side streets; 11 points (GM001–GM011) in Gaffar Manzil covering the residential interior and main access road; and 5 points (OH001–OH005) in Okhla Head along a transect perpendicular to Okhla Road. GPS coordinates (WGS84 decimal degrees) were recorded at each point using the smartphone's GPS module.

3.4 Monitoring sessions

Four sessions were conducted to capture temporal variation:

- Working Day Morning (WD-M): 10:40–13:09, Tuesday 03 March 2026 — Batla House.
- Working Day Evening (WD-E): Okhla Head 17:00–17:23 and Batla House 19:30–21:09, both on Wednesday 15 April 2026; Gaffar Manzil 20:00–20:59 on Tuesday 14 April 2026.
- Sunday Evening (SUN-E): Batla House and Okhla Head 19:30–21:41, Sunday 05 April 2026; Gaffar Manzil 19:30–20:30, Sunday 26 April 2026.
- Midnight (MID): 02:00–03:20, 26 April 2026.

3.5 Field protocol

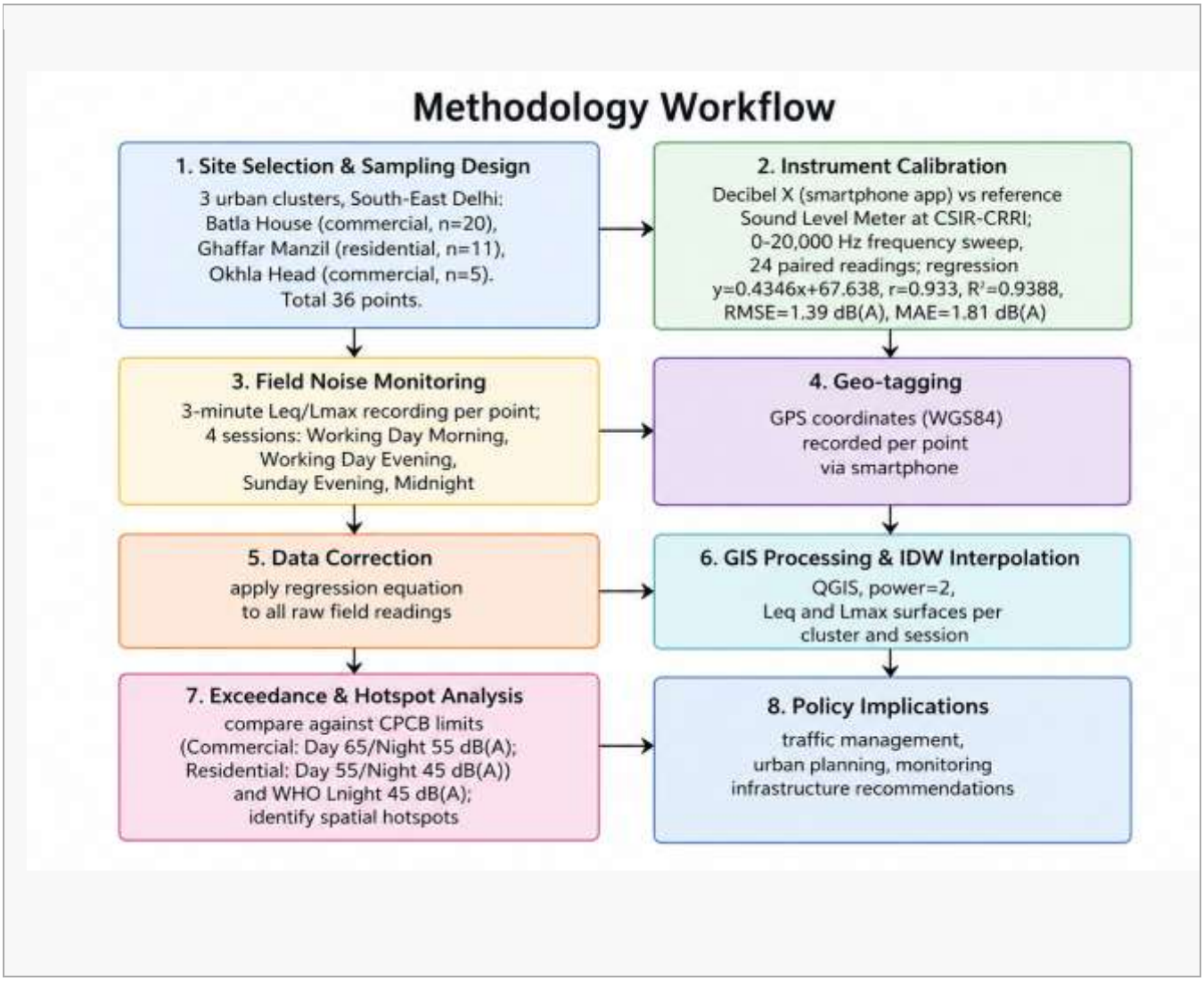
At each point, measurement ran continuously for three minutes, long enough to average short-duration fluctuating events (vehicle pass-bys, horn episodes, pedestrian activity) without compromising spatial sampling efficiency across 36 points per session. The device microphone was held approximately 1.0 m above ground and 0.5–0.6 m from the body. Wind speeds during all sessions were below 3 m/s; no meteorological correction was applied.

3.6 GIS spatial interpolation

Corrected noise data were imported into QGIS (version 3.x) as geo-referenced point layers, with Leq and Lmax attributed to each point. Spatial interpolation used the IDW algorithm (power = 2), selected over geostatistical methods such as ordinary Kriging because Kriging requires a sufficient number of points to fit a reliable variogram, which is not achievable for the 11-point (Gaffar Manzil) and 5-point (Okhla Head) datasets; IDW at this power setting is consistent with comparable Indian noise-mapping studies at similar point densities [14],[15],[16].

Interpolated surfaces were clipped to each cluster boundary and classified into five fixed classes using manual breaks, applied identically across all Leq maps (<45, 45–55, 55–65, 65–75, >75 dB(A)) and all Lmax maps (<70, 70–80, 80–90, 90–100, >100 dB(A)), using a consistent green–yellow–red colour scheme. Each output map includes a north arrow, scale bar, WGS84 coordinate reference, and the standardized legend, with basemap tiles sourced from OpenStreetMap.

Figure 3. Methodological workflow showing site selection and sampling design, instrument calibration, field noise monitoring, geotagging, data correction, GIS processing, IDW interpolation, exceedance analysis, hotspot identification, and policy implications



4. Results

4.1 Batla House (commercial zone)

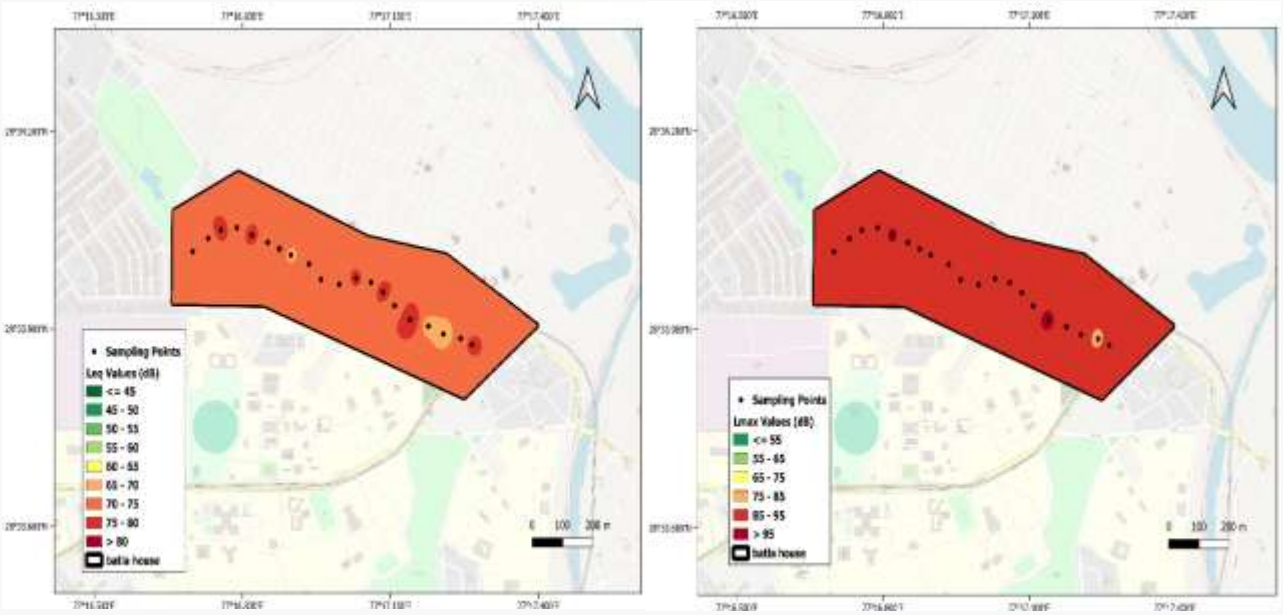
4.1.1 Working day morning (Tuesday 03 March 2026)

Twenty points were monitored between 10:40 and 13:09. Leq ranged from 68.2 to 80.5 dB(A), mean 72.9 dB(A); Lmax ranged from 82.0 to 97.5 dB(A), mean 90.7 dB(A). Against the CPCB commercial daytime limit of 65 dB(A), the mean Leq exceeded the standard by 7.9 dB(A).

Table 2

Batla House — working day morning data (03 March 2026).

Point ID	Time	Latitude	Longitude	Leq dB(A)	Lmax dB(A)
BH001	10:40	28.564601	77.28775	76.7	93.0
BH002	11:00	28.564759	77.287368	70.0	82.0
BH003	11:05	28.564869	77.286787	68.2	89.3
BH004	11:20	28.565068	77.286283	68.2	85.3
BH005	11:29	28.565224	77.285662	80.5	97.5
BH006	11:38	28.565589	77.285137	70.7	92.2
BH007	11:46	28.56593	77.284754	77.4	92.8
BH008	11:54	28.566178	77.28434	70.7	92.2
BH009	12:01	28.566278	77.283843	76.2	92.1
BH010	12:07	28.566127	77.283274	71.8	90.5
BH011	12:12	28.566251	77.28266	73.1	91.3
BH012	12:18	28.566642	77.282256	74.4	90.6
BH013	12:29	28.56687	77.281641	68.2	86.8
BH014	12:36	28.567025	77.281256	75.7	92.0
BH015	12:40	28.567188	77.280857	70.1	85.8
BH016	12:48	28.567374	77.28033	76.6	96.9
BH017	12:52	28.567562	77.279813	72.4	89.5
BH018	13:00	28.567508	77.279276	76.8	90.4
BH019	13:04	28.56729	77.278849	72.8	88.6
BH020	13:09	28.566957	77.278321	73.4	90.3



Figures 4–5. *Batla House working day morning IDW interpolation maps for Leq and Lmax (03 March 2026).*

4.1.2 Working day evening (Wednesday 15 April 2026)

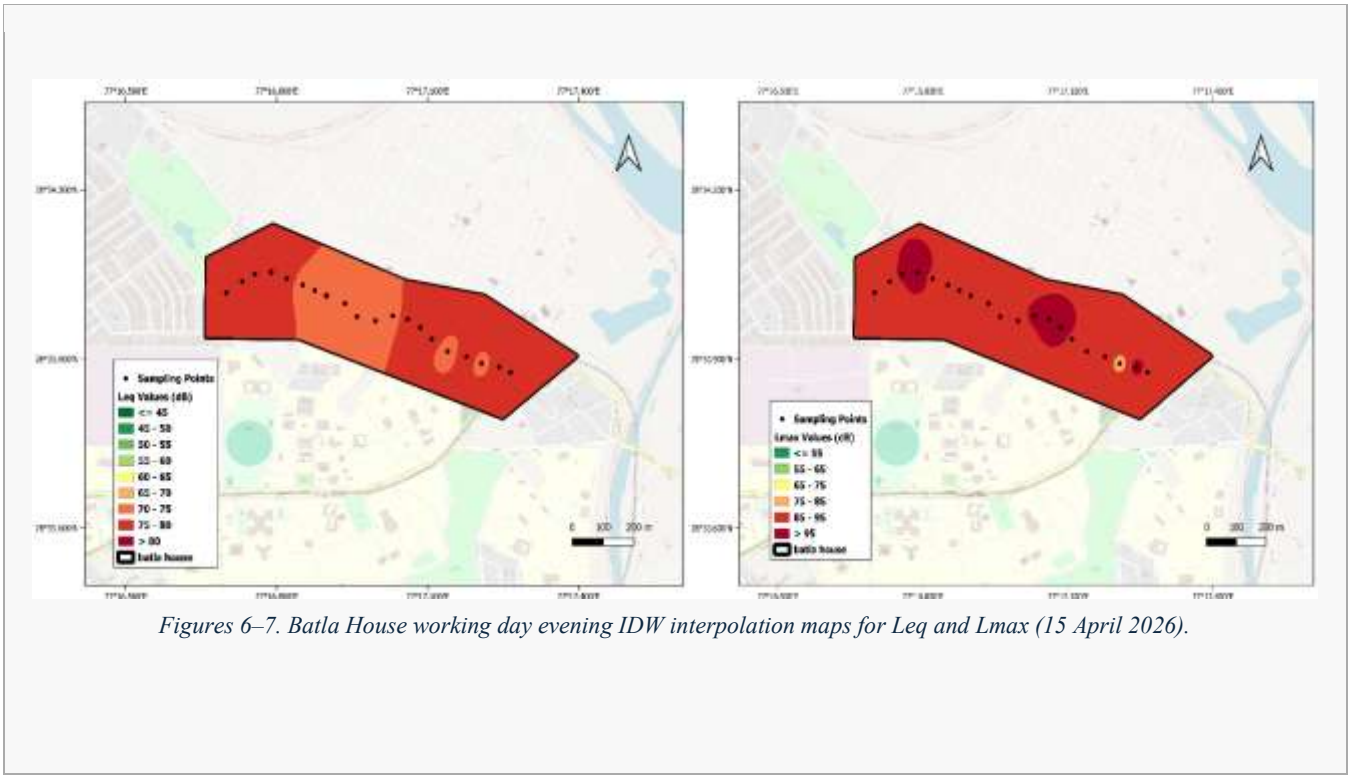
Evening monitoring produced the highest values recorded anywhere in this study. Leq ranged from 70.1 to 80.3 dB(A), mean 75.5 dB(A); Lmax reached 102.6 dB(A) at BH017, the highest single reading in the entire dataset, exceeding the CPCB commercial daytime limit of 65 dB(A) by 10.5 dB(A) on average.

Table 3

Batla House — working day evening data (15 April 2026).

Point ID	Time	Latitude	Longitude	Leq dB(A)	Lmax dB(A)
BH001	19:30	28.564601	77.28775	79.5	93.6
BH002	19:35	28.564759	77.287368	79.5	97.5
BH003	19:40	28.564869	77.286787	71.6	80.2
BH004	19:46	28.565068	77.286283	80.3	95.0
BH005	19:51	28.565224	77.285662	70.1	85.3
BH006	19:58	28.565589	77.285137	78.6	94.3
BH007	20:04	28.56593	77.284754	79.5	100.0
BH008	20:10	28.566178	77.28434	78.4	96.2
BH009	20:15	28.566278	77.283843	71.5	97.1
BH010	20:19	28.566127	77.283274	72.9	89.4
BH011	20:23	28.566251	77.28266	71.1	92.2
BH012	20:27	28.566642	77.282256	71.3	91.2
BH013	20:33	28.56687	77.281641	75.2	93.2
BH014	20:39	28.567025	77.281256	74.0	91.9
BH015	20:44	28.567188	77.280857	71.8	87.2
BH016	20:49	28.567374	77.28033	78.0	93.9

Point ID	Time	Latitude	Longitude	Leq dB(A)	Lmax dB(A)
BH017	20:55	28.567562	77.279813	80.3	102.6
BH018	21:00	28.567508	77.279276	77.3	96.2
BH019	21:04	28.56729	77.278849	80.0	88.9
BH020	21:09	28.566957	77.278321	75.0	91.0



Figures 6–7. Batla House working day evening IDW interpolation maps for Leq and Lmax (15 April 2026).

4.1.3 Sunday evening (05 April 2026)

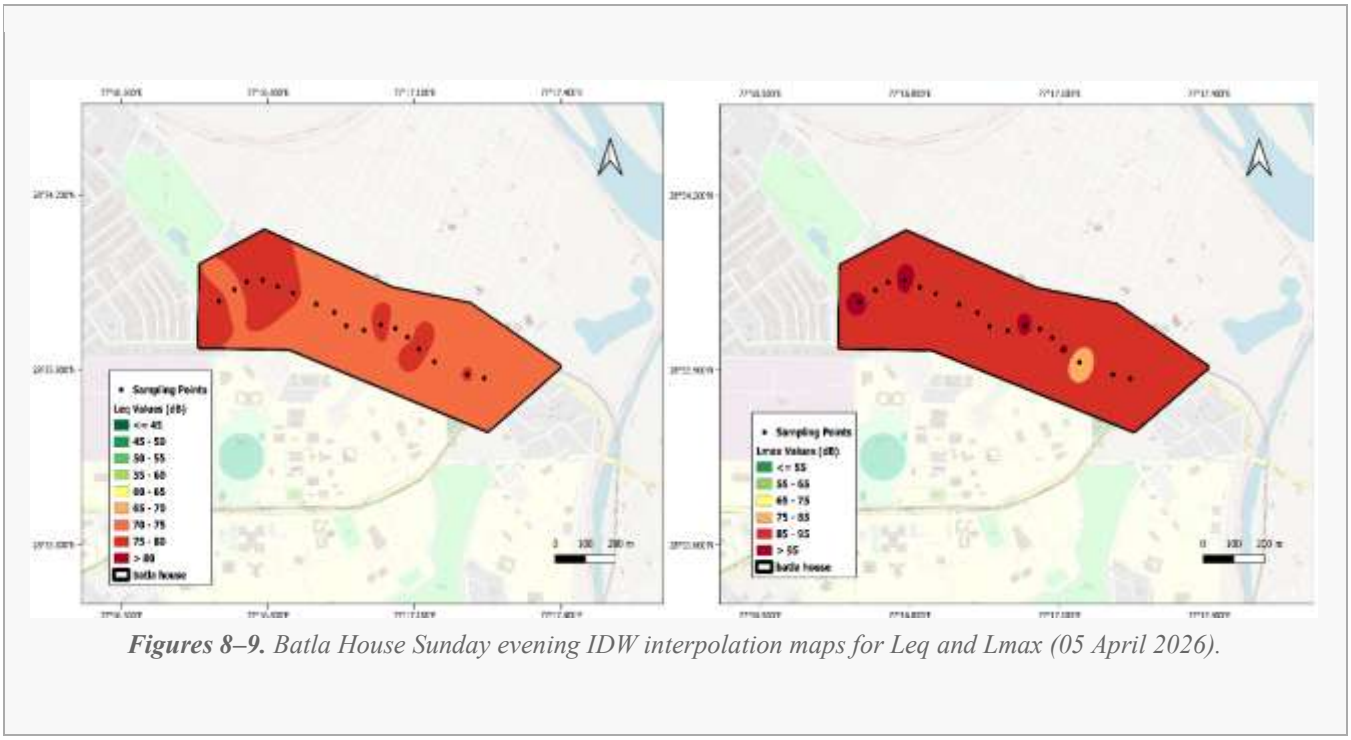
Sunday evening was almost as loud as a working day evening. Leq ranged from 70.7 to 80.4 dB(A), mean 74.7 dB(A), only 0.8 dB(A) below the working day evening mean. Lmax reached 101.3 dB(A) at BH017. This contrasts with Mahesh et al. [11], who reported 3–5 dB(A) Sunday reductions in Mysuru commercial zones.

Table 4

Batla House — Sunday evening data (05 April 2026).

Point ID	Time	Latitude	Longitude	Leq dB(A)	Lmax dB(A)
BH001	19:30	28.564601	77.28775	78.0	92.7
BH002	19:36	28.564759	77.287368	74.7	93.3
BH003	19:42	28.564869	77.286787	75.1	94.5
BH004	19:48	28.565068	77.286283	73.4	88.1
BH005	19:52	28.565224	77.285662	71.9	75.9
BH006	19:59	28.565589	77.285137	78.1	96.3
BH007	20:05	28.56593	77.284754	74.6	90.0
BH008	20:11	28.566178	77.28434	73.1	87.1
BH009	20:17	28.566278	77.283843	78.1	100.1
BH010	20:24	28.566127	77.283274	72.2	89.6
BH011	20:29	28.566251	77.28266	73.9	88.7

Point ID	Time	Latitude	Longitude	Leq dB(A)	Lmax dB(A)
BH012	20:36	28.566642	77.282256	71.4	86.7
BH013	20:41	28.56687	77.281641	70.7	87.1
BH014	20:50	28.567025	77.281256	71.8	88.9
BH015	20:59	28.567188	77.280857	75.9	93.0
BH016	21:06	28.567374	77.28033	76.4	91.3
BH017	21:11	28.567562	77.279813	75.0	101.3
BH018	21:18	28.567508	77.279276	80.4	88.0
BH019	21:25	28.56729	77.278849	70.0	85.0
BH020	21:30	28.566957	77.278321	76.5	97.7



Figures 8–9. *Batla House Sunday evening IDW interpolation maps for Leq and Lmax (05 April 2026).*

4.1.4 Midnight (26 April 2026, 02:00–03:20)

Mean Leq was 60.3 dB(A), exceeding the CPCB commercial nighttime limit of 55 dB(A) by 5.3 dB(A) and the WHO Lnight guideline of 45 dB(A) by 15.3 dB(A). Range was 53.3–74.4 dB(A); mean Lmax 77.2 dB(A). Dominant sources were heavy vehicles on adjacent Okhla Road, late-running auto-rickshaws, and occasional residual commercial activity.

Table 5

Batla House — summary by monitoring period (commercial zone limits).

Period	Mean Leq dB(A)	Mean Lmax dB(A)	CPCB commercial limit dB(A)	Exceedance
WD morning (03 March)	72.9	90.7	65 (day)	+7.9
WD evening (15 Apr)	75.5	93.6	65 (day)	+10.5
Sunday eve. (05 Apr)	74.7	91.8	65 (day)	+9.7
Midnight (26 Apr)	60.3	77.2	55 (night)	+5.3

4.2 Gaffar Manzil (residential zone)

4.2.1 Working day evening (Tuesday 14 April 2026)

Leq ranged from 52.1 to 74.0 dB(A), mean 66.3 dB(A); Lmax reached 95.2 dB(A) at GM011. Against the CPCB residential daytime limit of 55 dB(A), this exceeds the standard by 11.3 dB(A) — the largest daytime exceedance margin recorded in this study.

Table 6

Gaffar Manzil — working day evening data (14 April 2026).

Point ID	Time	Latitude	Longitude	Leq dB(A)	Lmax dB(A)
GM001	20:00	28.558154	77.283632	71.8	94.5
GM002	20:15	28.557574	77.283346	67.3	89.4
GM003	20:19	28.557283	77.283293	74.0	92.5
GM004	20:23	28.556591	77.283797	63.0	81.6
GM005	20:30	28.556204	77.284123	58.2	76.9
GM006	20:33	28.555489	77.284628	64.5	84.4
GM007	20:36	28.555067	77.284885	52.1	68.6
GM008	20:40	28.555135	77.28393	72.1	90.9
GM009	20:45	28.555716	77.283551	68.1	90.2
GM010	20:50	28.556221	77.283314	65.1	83.6
GM011	20:59	28.556655	77.282998	70.4	95.2

4.2.2 Sunday evening (26 April 2026)

Mean Leq was 64.9 dB(A), about 1.4 dB(A) below the working day evening figure, still exceeding the CPCB residential daytime limit by 9.9 dB(A).

Table 7

Gaffar Manzil — Sunday evening data (26 April 2026).

Point ID	Time	Latitude	Longitude	Leq dB(A)	Lmax dB(A)
GM001	19:30	28.558154	77.283632	67.6	90.9
GM002	19:35	28.557574	77.283346	70.0	91.2
GM003	19:40	28.557283	77.283293	68.6	89.0
GM004	19:48	28.556591	77.283797	62.5	79.4
GM005	19:55	28.556204	77.284123	62.6	77.8
GM006	19:59	28.555489	77.284628	60.2	72.5
GM007	20:05	28.555067	77.284885	55.8	69.5
GM008	20:11	28.555135	77.28393	72.2	95.0
GM009	20:17	28.555716	77.283551	66.3	86.6
GM010	20:24	28.556221	77.283314	66.9	88.2
GM011	20:30	28.556655	77.282998	67.5	90.0

4.2.3 Midnight (26 April 2026)

Mean Leq was 52.6 dB(A), the lowest of any cluster-session in the study, but still 7.6 dB(A) above the CPCB residential nighttime limit of 45 dB(A) and the WHO Lnight guideline.

Table 8

Gaffar Manzil — midnight data (26 April 2026).

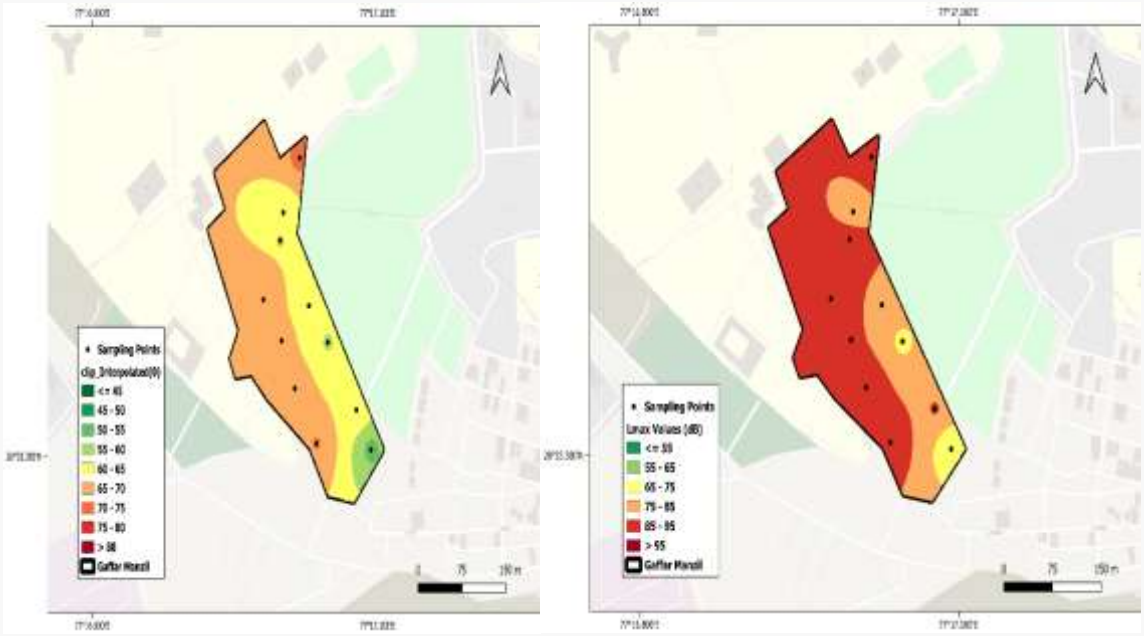
Point ID	Time	Latitude	Longitude	Leq dB(A)	Lmax dB(A)
GM001	02:00	28.558154	77.283632	48.7	62.1
GM002	02:08	28.557574	77.283346	53.8	70.9
GM003	02:19	28.557283	77.283293	56.6	79.6
GM004	02:23	28.556591	77.283797	56.6	79.2
GM005	02:30	28.556204	77.284123	50.6	69.5
GM006	02:33	28.555489	77.284628	48.9	71.0
GM007	02:36	28.555067	77.284885	49.4	61.6
GM008	02:40	28.555135	77.28393	53.9	72.7
GM009	02:45	28.555716	77.283551	55.1	74.7
GM010	02:50	28.556221	77.283314	54.3	61.1
GM011	02:59	28.556655	77.282998	56.3	75.9

Table 9

Gaffar Manzil — summary by monitoring period (residential zone limits).

Period	Mean Leq dB(A)	Mean Lmax dB(A)	CPCB residential limit	Exceedance
WD evening (14 Apr)	66.3	85.1	55 (day)	+11.3
Sunday eve. (26 Apr)	64.9	83.6	55 (day)	+9.9
Midnight (26 Apr)	52.6	70.4	45 (night)	+7.6

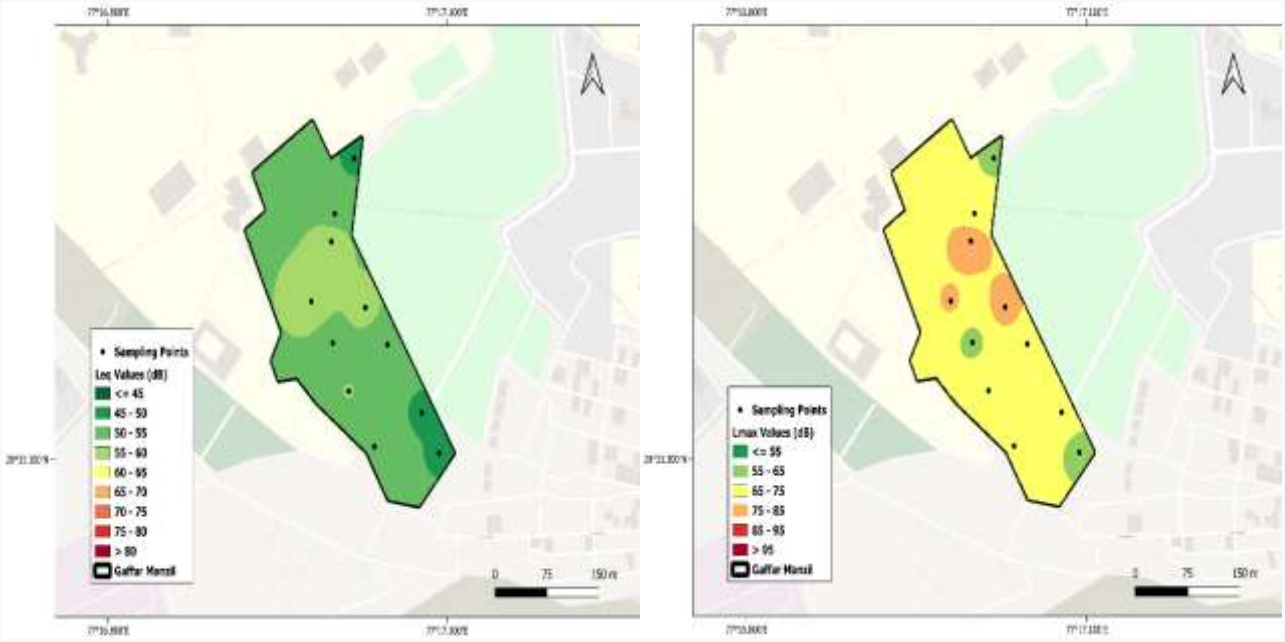
Fig. 10. Gaffar Manzil — IDW spatial interpolation maps for all monitoring sessions (Leq and Lmax).



Figures 10–11. *Gaffar Manzil working day evening IDW interpolation maps for Leq and Lmax (14 April 2026).*



Figures 12-13. *Gaffar Manzil Sunday Evening IDW Interpolation maps for Leq and Lmax (26 April 2026)*



Figures 14–15. *Gaffar Manzil midnight IDW interpolation maps for Leq and Lmax (26 April 2026).*

4.3 Okhla Head (commercial zone, arterial corridor)

Okhla Head is acoustically simpler than the other two clusters: one dominant source, Okhla Road traffic, drives most of the noise, with the IDW maps showing highest levels at OH005 (closest to the road) declining towards OH001.

4.3.1 Working day evening (Wednesday 15 April 2026)

Five points monitored 17:00–17:23. Leq ranged from 68.0 to 73.6 dB(A), mean 71.4 dB(A), exceeding the CPCB commercial daytime limit of 65 dB(A) by 6.4 dB(A).

4.3.2 Sunday evening (05 April 2026)

Mean Leq was 72.2 dB(A), marginally higher than the working day figure, exceeding the commercial daytime limit by 7.2 dB(A).

4.3.3 Midnight (26 April 2026)

Mean Leq was 59.6 dB(A), exceeding the CPCB commercial nighttime limit of 55 dB(A) by 4.6 dB(A) and the WHO Lnight guideline by 14.6 dB(A). OH005 recorded Leq 69.4 dB(A) at 02:27, the highest midnight reading among the five points, attributable to heavy goods vehicles dominating Okhla Road traffic composition at that hour.

Table 10

Okhla Head — full dataset with GPS coordinates.

ID	Session	Date	Latitude	Longitude	Leq	Lmax	Exceedance
OH001	WD eve	15/4/2026	28.56239	77.28660	68.0	84.6	+3.0
OH002	WD eve	15/4/2026	28.56232	77.28754	70.1	81.2	+5.1

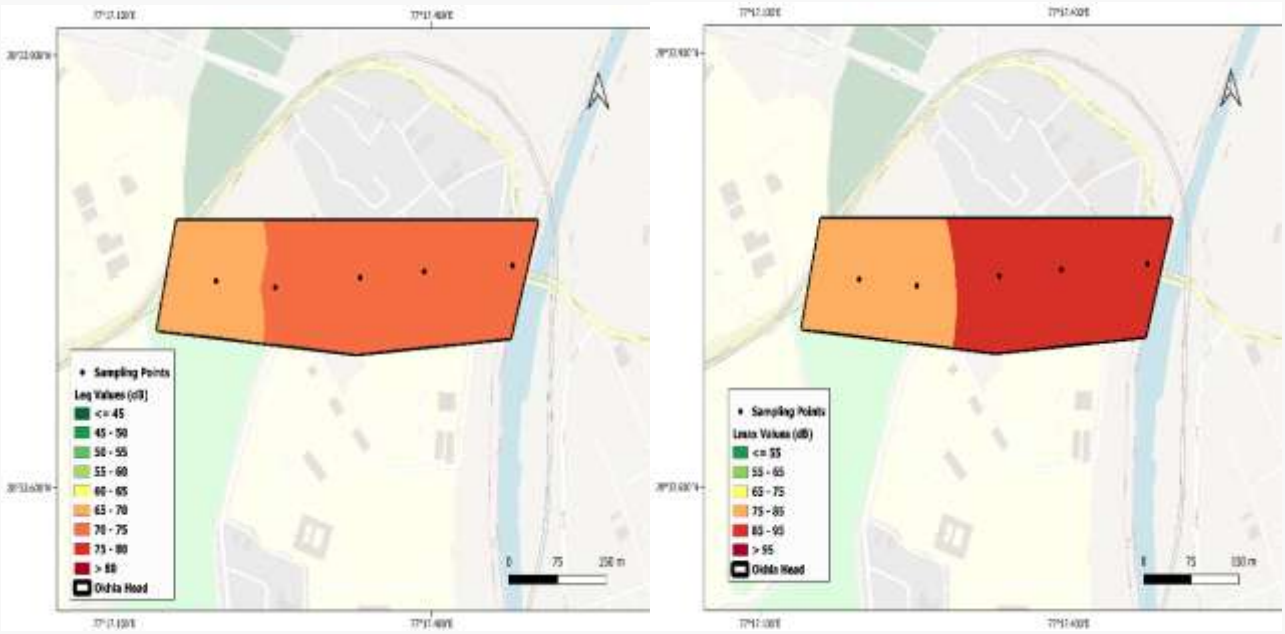
ID	Session	Date	Latitude	Longitude	Leq	Lmax	Exceedance
OH003	WD eve	15/4/2026	28.56243	77.28887	73.6	88.7	+8.6
OH004	WD eve	15/4/2026	28.5625	77.28988	72.1	91.6	+7.1
OH005	WD eve	15/4/2026	28.56257	77.29128	72.6	85.0	+7.6
OH001	Sun eve	05/4/2026	28.56239	77.28660	69.9	85.0	+4.9
OH002	Sun eve	05/4/2026	28.56232	77.28754	73.0	89.2	+8.0
OH003	Sun eve	05/4/2026	28.56243	77.28887	74.9	92.9	+9.9
OH004	Sun eve	05/4/2026	28.5625	77.28988	74.9	94.6	+9.9
OH005	Sun eve	05/4/2026	28.56257	77.29128	73.0	87.9	+8.0
OH001	Midnight	26/4/2026	28.56239	77.28660	57.0	70.4	+2.0
OH002	Midnight	26/4/2026	28.56232	77.28754	56.2	67.2	+1.2
OH003	Midnight	26/4/2026	28.56243	77.28887	57.2	79.7	+2.2
OH004	Midnight	26/4/2026	28.5625	77.28988	58.3	70.6	+3.3
OH005	Midnight	26/4/2026	28.56257	77.29128	69.4	88.3	+14.4

Table 11

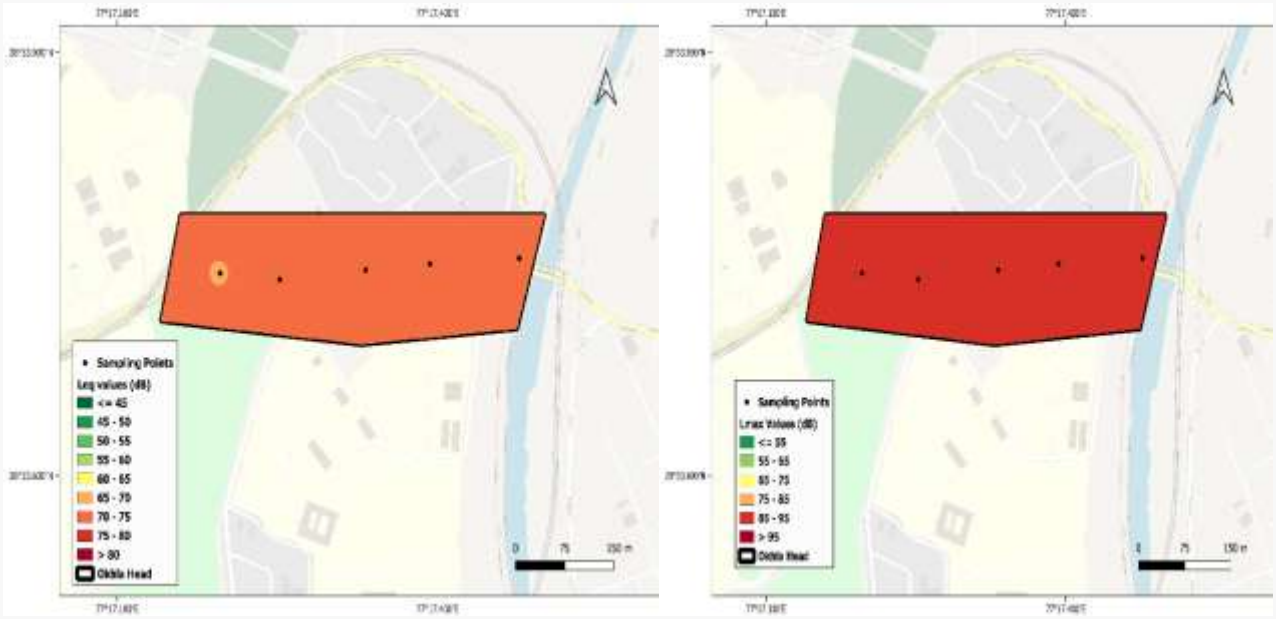
Okhla Head — summary by monitoring period (commercial zone limits).

Period	Mean Leq dB(A)	Mean Lmax dB(A)	CPCB commercial limit	Exceedance
WD evening (15 Apr)	71.4	86.6	65 (day)	+6.4
Sunday eve. (05 Apr)	72.2	89.4	65 (day)	+7.2
Midnight (26 Apr)	59.6	74.8	55 (night)	+4.6

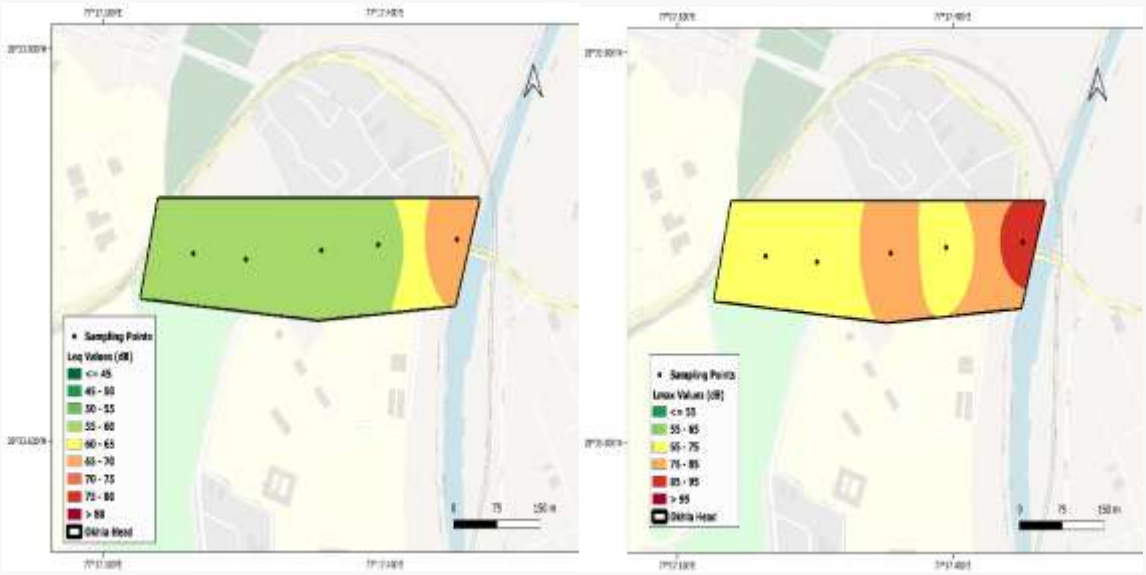
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Working Day Evening Leq vs Lmax



Sunday Evening Leq vs Lmax



Mid Night Leq Vs Lmax

Figure 16. Okhla Head IDW interpolation maps showing the spatial pattern of Leq and Lmax across the monitoring sessions.

4.4 Comprehensive exceedance analysis

Table 12

Full CPCB and WHO exceedance summary — all clusters and sessions.

Location (zone)	Period	Obs. Leq dB(A)	CPCB limit	Exceedance	WHO Lnight (midnight only)
Batla House (Comm.)	WD morn	72.9	65 (day)	+7.9	—
Batla House (Comm.)	WD eve	75.5	65 (day)	+10.5	—
Batla House (Comm.)	Sun eve	74.7	65 (day)	+9.7	—
Batla House (Comm.)	Midnight	60.3	55 (night)	+5.3	+15.3
Gaffar Manzil (Res.)	WD eve	66.3	55 (day)	+11.3	—
Gaffar Manzil (Res.)	Sun eve	64.9	55 (day)	+9.9	—
Gaffar Manzil (Res.)	Midnight	52.6	45 (night)	+7.6	+7.6
Okhla Head (Comm.)	WD eve	71.4	65 (day)	+6.4	—
Okhla Head (Comm.)	Sun eve	72.2	65 (day)	+7.2	—
Okhla Head (Comm.)	Midnight	59.6	55 (night)	+4.6	+14.6

Table 12 demonstrates that CPCB noise standards, applying the limit appropriate to each cluster's designated zone, are exceeded at every monitoring location during every monitoring period without exception. The magnitude of exceedance ranges from +4.6 dB(A) (Okhla Head midnight, commercial nighttime limit) to +11.3 dB(A) (Gaffar Manzil working day evening, residential daytime limit). Midnight exceedances against the WHO Lnight guideline of 45 dB(A) [1] range from +7.6 to +15.3 dB(A) across the three clusters, exceeding the 50–55 dB(A) sleep-disturbance threshold reported by Skånberg and Öhrström [28] at all three sites.

5. Discussion

5.1 Spatial acoustic hotspots

The GIS maps identify three recurring hotspot categories. The Batla House commercial corridor intersections (BH005, BH007, BH016–BH018) consistently record the highest Leq values in the study, where high traffic volume, frequent horn use, and narrow road width combine with three- to five-storey flanking buildings; the Lmax maps for working day periods show nearly the entire cluster above 80 dB(A). The Gaffar Manzil northern access corridor (GM001, GM008, GM011) consistently records the cluster's highest values, with Leq dropping by 15–20 dB(A) towards the quieter interior (GM007); a single IDW surface necessarily smooths across this access-road-to-interior transition rather than treating it as two distinct zones, a limitation discussed further in Section 6. The Okhla Head eastern transect (OH004–OH005) drives the cluster's highest readings, including the midnight outlier of 69.4 dB(A) at OH005.

5.2 Temporal patterns

Working day evening was the loudest session across all three clusters (Batla House 75.5 dB(A); Okhla Head 71.4 dB(A); Gaffar Manzil 66.3 dB(A)), reflecting the simultaneous peak of commercial activity, pedestrian crowding, and vehicular traffic. The Sunday evening comparison is notable: the reduction from working day evening was only 0.8 dB(A) at Batla House and 1.4 dB(A) at Gaffar Manzil, with a marginal increase at Okhla Head — considerably smaller than the 3–5 dB(A) weekend reductions reported in Mysuru commercial zones [11], reflecting Batla House's function as a weekend leisure and retail destination rather than a purely weekday commercial corridor. Midnight brought the largest attenuation but left all sites above both CPCB and WHO limits.

5.3 Environmental health implications

Chronic exposure above 55–60 dB(A) Leq is associated with elevated risk of ischaemic heart disease and arterial hypertension [27]. Nighttime exposure above 45 dB(A) causes sleep fragmentation, cortisol elevation, and cardiovascular stress responses even without subjective awareness of waking [1]. Das et al. [12] similarly paired noise exposure mapping with vulnerability assessment in Eastern India, underlining how exposure data of the kind generated here can be linked to population-level health risk in future work. The presence of hospitals, mosques, and schools within and immediately adjacent to the study area means that silence zone standards, which require noise levels 5 dB(A) below normal limits near sensitive receptors, are almost certainly being violated, though their formal designation was not verified in this study.

5.4 Policy implications

These findings support several prioritized interventions. For traffic and enforcement: designating the Batla House commercial corridor core as a no-honking zone with active enforcement; vendor relocation and no-parking enforcement to reduce stop-start congestion; and pothole repair and traffic-calming measures. For urban planning: acoustic setback and insulation requirements for new construction in mixed land-use zones, particularly Gaffar Manzil, where the residential classification does not reflect actual ambient exposure; restrictions on ground-floor commercial use abutting residential streets; and green buffer planting along Okhla Road and Batla House Road for co-benefit noise attenuation. For monitoring infrastructure: a continuous fixed-sensor network at the identified hotspots; calibrated smartphone-based citizen monitoring as a low-cost supplementary tool for other under-monitored clusters in Delhi NCR; and six-monthly public ambient noise reporting for Jamia Nagar.

6. Limitations

- The monitoring covered four discrete sessions rather than continuous measurement, and cannot capture seasonal variation, festive period noise events, or long-term trends.

- All sessions fall within a single year; monsoon-season conditions, which typically reduce traffic and alter propagation, are absent from the dataset.
- Wind speeds were below 3 m/s during all sessions and no meteorological correction was applied; temperature gradient and humidity effects were not characterised.
- Calibration was conducted with a single smartphone unit using a 0–20,000 Hz frequency sweep played through a laptop speaker; device-to-device microphone variability was not assessed, and the calibration stimulus differs acoustically from real traffic noise.
- Sampling density, while adequate for cluster-scale IDW mapping, limits resolution within the narrowest interior lanes. As discussed in Section 1.1, IDW interpolation does not account for the spatial heterogeneity that arises when adjacent areas have markedly different land-use-driven noise regimes; a region-stratified interpolation approach such as that proposed by Zuo et al. [22] could improve map accuracy in future work given a larger sample size.
- The study quantifies acoustic exposure but does not separate contributions from individual source types; source apportionment would enable more targeted mitigation design.

7. Future research directions

- Deployment of IoT-enabled fixed noise sensors at identified hotspots for continuous, multi-season monitoring.
- Application of machine learning classifiers to acoustic recordings for automated noise source identification.
- Citizen-science smartphone monitoring to achieve wider spatial coverage following validated calibration protocols.
- With larger sample sizes, application of region-stratified interpolation methods [22] and formal hotspot statistics to better account for spatial heterogeneity between adjacent land-use types.
- Linkage of the geo-referenced exposure dataset to measured health outcomes (blood pressure, sleep quality, annoyance) in the resident population.

8. Conclusion

This study produced a systematic, geo-referenced, multi-temporal acoustic dataset for three high-density urban micro-clusters in Jamia Nagar, South-East Delhi, using a smartphone (Decibel X) validated against a reference Sound Level Meter at CSIR–CRRI ($r = 0.933$, $R^2 = 0.9388$, RMSE = 1.39 dB(A), MAE = 1.94 dB(A)). Applying zone-appropriate CPCB limits, noise standards were exceeded at every cluster during every monitoring period, with exceedance margins ranging from 4.6 to 11.3 dB(A). Gaffar Manzil, despite its residential classification, recorded the largest daytime exceedance; Batla House recorded the highest absolute levels, with working day evening Leq of 75.5 dB(A) and Lmax peaking at 102.6 dB(A). At midnight, all three clusters exceeded the WHO Lnight guideline of 45 dB(A), indicating measurable risk of sleep disruption for residents. GIS-based IDW mapping identified persistent hotspots at commercial frontages, arterial road intersections, and main access corridors, providing spatially specific evidence for noise governance. The calibrated smartphone–GIS methodology demonstrated here is scalable and low-cost, and is readily extendable to other under-monitored high-density urban clusters across Delhi and other Indian cities.

. Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this manuscript, the author used Claude (Anthropic) to assist with language editing, content organization, and structuring of the methodology and results sections, and to support consistency checks across statistical reporting. All research design, data collection, calibration, GIS analysis, and scientific

interpretation were performed independently by the author. The author has reviewed, verified, and takes full responsibility for all data, analysis, and conclusions presented in this publication.

. Declaration of competing interest

The author declares no competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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. Data availability

The geo-referenced noise monitoring dataset generated and analysed in this study is available from the corresponding author upon reasonable request.

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